ROPE ACCESS, BODY HARNESS, PFD

PURPOSE:

This Standard Operating Procedure Rope Access Body Harness is to identify potential hazards and outline the steps to Don a Rope Access Body Harness.

HAZARDS

TYPE	HAZARD	TYPE	HAZARD	TYPE	HAZARD
Physical	Slips, Trips, Falls	**Chemical**	H2S	**Property**	Equipment
	Fall from Height		Fuel		Third Party Property
	Pinch Points		Cement	**Productivity**	Down time
	Heated/Hot Surface	**Biological**	Wildlife		Inefficient
	Noise 82<		Insect bites	**Legal**	Suspension
	Energy		Influenza		Out of Service
	Environment	**Social**	Harassment		Fine
	Confined Space		Violence		
	Work Alone				

REFERENCES

PPE	Prevention	References
Fire Retardant Coveralls	Hazard Assessment	Alberta OH&S
Steel Toed Boots	Inspection	Task List
Safety Glasses	Maintenance	Standard Operating Procedure
Gloves	FLHA	
Hard Hat	Safety Meeting	
Hearing Protection	Ear Plugs / Muffs	

You are required to perform at a Job, there are some items that need to be considered:	
Is this your first job, task or step;	Do you have a Safety Attendant;
Has the Job/Task been Risk Assessed;	Will the Safety Attendant be accompanying you;
How far away is the Work Site, Travel Time, ERP Response Time;	Will the Safety Attendant be available by cell phone or other;
Do you have a WORK ALONE PLAN in place;	
Do you have a Rope Access Plan in place; Learn More	

BODY HARNESS & LIFELINE

When the job does require the worker to be connected to a lifeline, i.e., for extraction when the worker becomes unable to help him or herself:
A **Rope Access Safe Work Plan** must be implemented:

Rope Access: When conducting work requiring rope access, a special type of fall protection plan (called a "rope access safe work plan") is needed. The requirements for this rope access safe work plan are specified in Section 809 of the OHS Code. Section 809 is under Part 41 of the OHS Code, which contains all the requirements pertaining to work where rope access is needed.

When a space is restricted, a body of water or other space poses a significant risk, a Hazard Management Plan and Rescue Plan must be in place, including a Safety Attendant, a harness, and a lifeline. Understanding that your workspace may be a danger for others to lend assistance quickly, is why a lifeline connected to a harness worn by the worker in trouble, would allow the worker to be pulled to safety.

CONFINED SPACE: will require a number of requirements to be addressed. Silverpoint does not perform work in a Confined Space. If the workspace is a confined space, **Stop Work**, be sure to talk to your Supervisor.

Wearing a harness correctly can save you when you are working in a restricted space:

1) you loose stability traction or other;
2) are unable to remove yourself under your own power;
3) you receive a disabling injury.

Working in a restricted space, on Ice and/or a elevated platform where fall retention is required. while wearing a harness with a lifeline long enough to be "Tied Off" at a safe location would allow:

1) you to be able to pull yourself to safety;
2) another worker (Safety Attendant) to assist in pulling you to safety.

Consider the environment, time for help to arrive, and the presence of wildlife in making your decision to have a Safety Attendant by your side or have one within calling distance.

NOTE: This is not a "Fall Arrest Plan" this is a "Rope Access Safe Work Step" it is important to know the difference and what is applicable in which type of work situations. If there is danger of falling from an elevated work platform a "Fall Arrest Plan" must be implemented

The following is a general procedure for putting on ("Donning") a harness (Rope Access):

INSPECT THE HARNESS: Before each use, inspect the harness for any signs of wear, damage, or deterioration. Check for frayed edges, broken fibers, pulled stitches, cracks, and any other signs of damage or excessive wear in the straps, buckles, and attachment points.

LAY OUT THE HARNESS: Lay the harness out on a flat surface to identify all its straps (shoulder straps, leg straps, chest strap, and waist belt if applicable) and to make sure the straps are not twisted.

DON THE HARNESS: Hold the harness by the D-ring located on the back of the harness. Locate the chest strap. Slip one arm through one side of the harness, and the other arm through the other side; the same as you would putting a jacket on. The D-ring should be centered in the middle of your back just below your neck. When doing the harness up, always be sure to start with the leg straps. Next, buckle the chest strap. This strap should sit at mid-chest level and be snug to the chest cavity. The chest strap is used to keep the shoulder straps of the harness from slipping off. Remember to always follow the manufacturer's instructions for putting on a harness.

Watch the training video in the Work Hub Online training selection **How to Don a Body Harness.**

ADJUST THE STRAPS: Adjust the leg, chest, shoulder, and waist straps (if applicable) to ensure a snug but comfortable fit. The harness should be tight enough, so it does not slip, but not so tight as to be uncomfortable or restrict movement.

CHECK FIT: After adjusting, you should not be able to fit more than 2 or 3 fingers snuggly between the strap and your body. At the same time, ensure you have full range of motion. It is often helpful to have a colleague check that everything is adjusted correctly.

CONNECT TO A LIFELINE: Once the harness is on and adjusted, connect it to an inspected and certified lifeline which is then connected to a secure tie off point. Ensure that all connections are secure, and the locking mechanisms of carabiners or hooks are engaged.

REGULAR CHECKS: Periodically check the harness and the connections to the lifeline or lanyard throughout the job to ensure they remain secure and properly positioned.

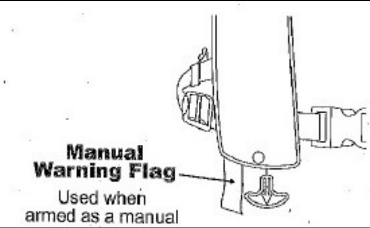
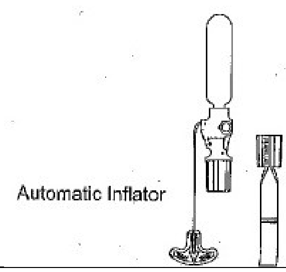
The exact method can vary slightly depending on the specific design of the harness. Always follow the manufacturer's instructions for your specific harness model. Additionally, if you're new to using a fall arrest harness, it's highly recommended to receive proper training from a qualified professional.

IMPORTANT:

The Harness must be worn underneath the Personal Flotation Vest.

PERSONAL FLOTATION DEVICE

Working close to water is a hazardous part of a job, the purpose of a personal flotation device (PFD) on a job site, particularly in environments near or on water, is primarily for safety. PFDs are designed to keep an individual afloat in the event of an accidental fall into water, thereby reducing the risk of drowning.

Don a Personal Flotation Device	
Check the inflation system status indicator	
Automatic Inflator	

Donning a flotation vest, also known as a personal flotation device (PFD), properly is crucial to ensure its effectiveness in providing buoyancy and keeping you safe in the water. Here are the general steps to don a flotation vest:

Select the Right Size: Ensure you have the correct size of PFD that fits you snugly. There are different sizes available, so choose the one that suits your body size.

Inspect the PFD: Before putting it on, inspect the PFD for any signs of damage, wear, or malfunction. Check the straps, buckles, and any other fasteners to make sure they are in good condition.

Open the Straps: If the PFD has multiple straps, open them all before attempting to put it on.

Hold the PFD in Front of You: Hold the PFD with both hands, one on each side, with the front of the PFD facing you.

Slip Arms Through the Arm Openings: Insert your arms through the arm openings of the PFD. The PFD should now rest on your shoulders with the front of the PFD on your chest and the back on your back.

Secure the Front Closure: Depending on the type of PFD, secure the front closure. This could be a zipper, buckles, or straps. Make sure it is snug but not too tight, allowing for comfort and mobility.

Adjust the Straps: If your PFD has adjustable straps, tighten or loosen them as needed to ensure a secure fit. Straps should be snug, but not so tight that they restrict your movement or breathing.

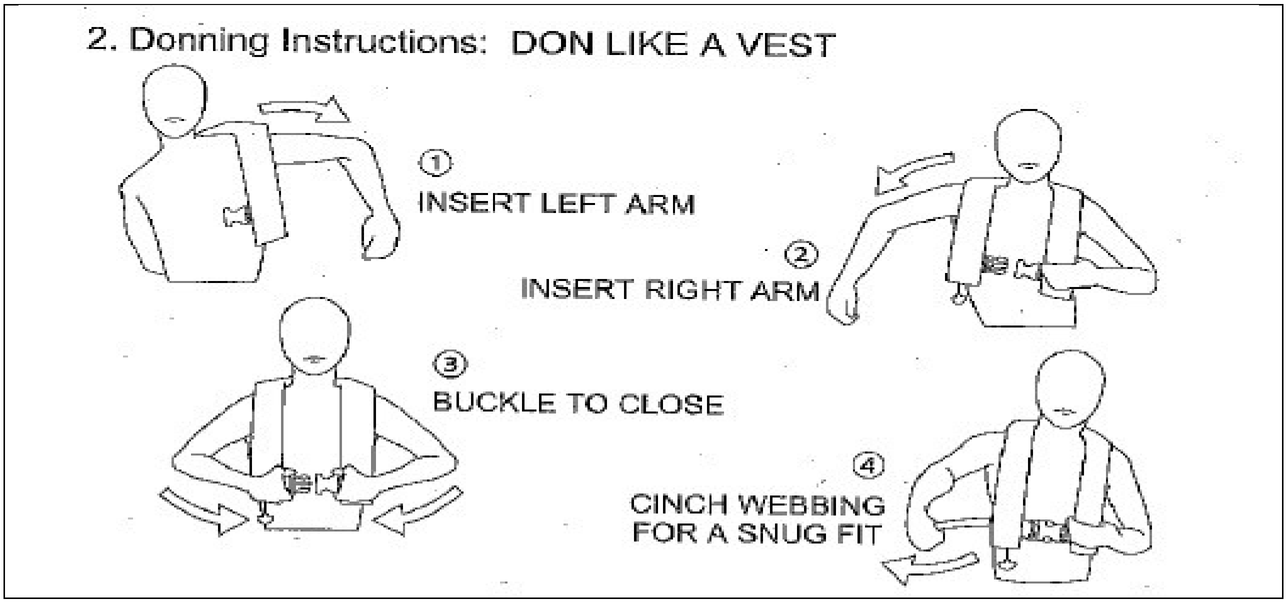
Check for a Proper Fit: Ensure that the PFD fits snugly around your torso. It should not ride up or be so loose that it can slip off.

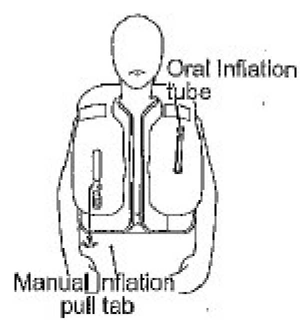
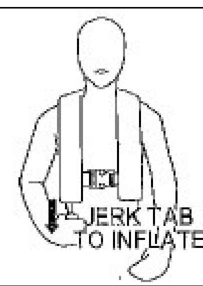
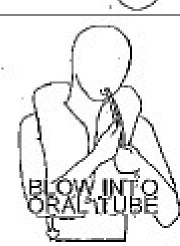
Secure Additional Straps: If there are any additional straps on the PFD, such as thigh or crotch straps, secure them according to the manufacturer's instructions. These straps help prevent the PFD from riding up during use.

Ensure Proper Positioning: Make sure the PFD is properly positioned on your body, with the buoyant material covering your chest and upper back.

Check for Comfort and Mobility: Move your arms and torso to ensure that the PFD allows for comfortable movement while still providing proper buoyancy.

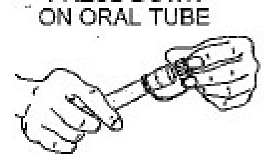
Remember, the specific steps may vary depending on the design and type of PFD. Always refer to the manufacturer's instructions for the particular flotation deviceyou are using, and follow any guidelines provided for proper fitting and use. Regularly inspect and maintain your PFD to ensure it remains in good working condition.



Inflating the Personal Flotation Vest	
CO2 Inflation (Automatic/Manual PFD) The PFD will inflate when fully immersed in water. Automatic/Manual PFDs can be manually inflated by grasping the manual inflation pull tab and pulling sharply downward.	
CO@ inflation (Manual Inflation PFD) the PFD is manually inflated by grasping the manual inflation pull tab and pulling sharply downward.	
Oral inflation the Oral inflation tube is located inside the cover on the wearers left side. To inflate orally access the inflation tube by opening the top portion of the cover remove the dust cap and blow air into the tube until the inflatable is firm.	

NOTE: Oral Inflation can be used to test for air leakage or as a backup in the event that the CO2 cylander fails to operate. CO2 gas permeates the bladder more quickly than air and therefore oral inflation can be used to refill the bladder in the event the wearer is in the water for an extended period.

The deflation tab is located on top of the dust cap Turn the dust cap upside down and insert the deflation mechanism into the oral tube. Hold the deflation mechanism in place with your finger and gently squeeze all the air out of the PFD. If the PFD has inflated while being worn it may be necessary to partially deflate the PFD in order to remove it.



PRACTICE wearing and adjusting your inflatable PFD in the water in order to be prepared for a emergency.
Practice wearing and inflating your PFD using different types of clothing Wade into the water and manually inflate or allow your PFD to inflate auto-matically. Practise until you have complete confidence in the performance of our PFD.

LETS TALK HYPOTHERMIA

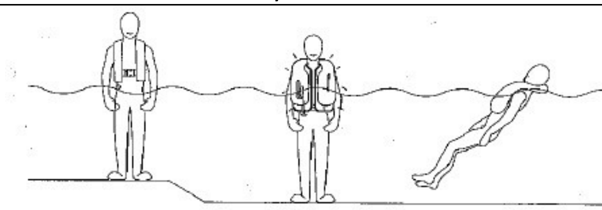
Prolonged exposure to Cold water causes a condition known as hypothermia – a substantial loss of body heat which leads to exhaustion and unconsciousness. Most drowning victims first suffer from hypothermia The following chart shows the effects of hypothermia

How Hypothermia Affects Most Adults		
WATER TEMPERATURE C (F)	**Exhaustion or Unconsciousness**	**Expected Time of Survival**
0.3 (32.5)	Under 15 Min	Under 15 to 45 Min
0.3 to 4 (32.5 to 40)	15 to 30 Min	30 to 90 Min
4 to 10 (40 to 50)	30 to 60 Min	1 to 3 Min
10 to 16 (50 to 60)	1 to 2 Hours	1 to 6 Hours
16 to 21 (60 to 70)	2 to 7 Hours	2 to 40 Hours
21 to 27 (70 to 80)	2 to 12 Hours	3 Hours to Indefinite
Over 27 (over 80	Indefinite	Indefinite

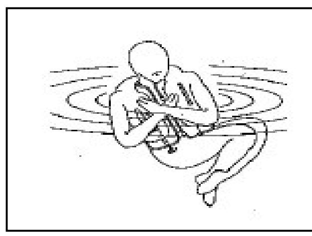
PERSONAL FLOTATION DEVICE

PFDs can increase survival time because they allow you to float without using energy treading water and because of their insulating properties. Naturally, the warmer the water, the less insulation you will require. When operating in cold water (below 60 F (15.6 C) you should consider using a coat or jacket style PFD or a Type V Thermal Protective PFD as they cover more of the body than the device or belt style PFDs.
Some points to remember about Hypothermia Protection:

WHY ARE PFDs REQUIRED SAFETY EQUIPMENT?
Drownings are the leading cause of fatalities involving recreational boating. A PFD provides flotation to help keep your head above water, help you to stay · face up in the water, and increase your chances for survival and rescue. Most adults only need an extra 7 to 12 pounds of flotation to keep their heads above water. The proper size PFD will properly support the weight of the wearer.
Since this inflatable PFD does not have inherent buoyancy, it provides inflation only when inflated. Familiarize yourself with the use of this PFD so you know what to do in an emergency.



1. **Always wear you PFD:** Even if you become incapacitated due to hypothermia, the PFD will keep you afloat and greatly improve your chances of rescue.
2. **Do not attempt to swim:** unless it is to reach a nearby craft, fellow survivor, or a floating object on which you can lean or climb. Swimming increases the rate of body heat loss. In cold water, drown-proofing 'methods that require putting your head in the water are not recommended. Keep your head out of the water. This will lessen heat loss and increase your survival time.



3. **Use the standard H.E.L.P. position:** when wearing an inflatable PFD, drawing the legs up to a seated position, because doing so will help you conserve body heat.
4. **Keep a positive attitude:** about your survival and rescue. This will improve your chances of extending your survival time until rescued Your will-to-live does make a "difference!
5. **If, there is more than one person in the water:** huddling is recommended while waiting to be rescued. This action tends to reduce the rate of heat loss and thus increase the survival time.

SELF RESCUE

ASSESS THE SITUATION: If you find yourself in a situation where self-rescue is necessary and achievable, the first step is to stay calm and assess your situation. Check for any immediate dangers, such as unstable structures or falling debris, as well as potential obstacles in your path.

COMMUNICATION: If possible, communicate with your Safety Attendant and/or emergency services, explaining your situation.

EXTRACT YOURSELF; WATER: If you fell into water, broke through ice and into the water, remain calm:

- **a. Inflate your Personal Flotation Vest;**
- **b. Call for help;**
- **c. Assess your situation:**
    - **1. How deep is the water? Can you stand?**
    - **2. How far away from land are you?**
    - **3. Is your lifeline tight between you and the tie off point?**

Depending on your situation, decide whether you need to extract, ascend to a safe point, or descend to the ground or another safe spot. Plan your route carefully, taking into consideration any potential obstacles.

EXTRACT YOURSELF; ASCEND OR DESCEND, If you need to escape your space:

- **a. Remain calm;**
- **b. Call for help;**
- **c. Assess your situation:**
    - **1. How far are you from a Safe Location?**
    - **2. Do you need to exit horizontally, ascend or descend?**
    - **3. Is your lifeline tight between you and the tie off point?**

Depending on your situation, decide whether you need to extract, ascend to a safe point or descend to the ground or another safe spot. Plan your route carefully, taking into consideration any potential obstacles.

SELF RESCUE TECHNIQUES: Self-rescue techniques may include but are not limited to gentle twisting or turning and maneuvering of your body by crawling or scooting backwards to free yourself. Using your lifeline, a hand over hand process pulling your self closer to where the lifeline is tied off.
Remember to keep your movements deliberate and controlled, quick panicked movements could worsen the situation. As always remember to stay calm and relaxed as best that you can; this could help you get through a tight spot more easily!

CALL FOR ASSISTANCE: If you are not able to free yourself, call for assistance using communication devices or hand signals.

AWAIT RESCUE: There are a few things that you can do during and awaiting rescue. Be sure to communicate with your team. Provide them with details about your location, the nature of the problem and any immediate concerns. Assess your equipment and ensure that it is functioning correctly. If there is a problem be sure to communicate this with your team. Remain attached to your rope system and maintain a comfortable position. Avoid making any unnecessary movements that could compromise your safety. Be aware of your physical condition. Adjust your harness or position if discomfort or pain arises. If you are injured or have any medical conditions, be sure to communicate this with your team as well. Use any available signalling devices (flashlight, whistle, etc.) to make yourself more visible to rescuers. Follow all instructions that rescue personnel, or your team give you. Above all stay positive and focused. Being rescued could take some time, and maintaining a calm mindset is crucial for your mental and physical well-being.

SAFETY ATTENDANT RESCUE

ASSESS THE SITUATION: If an individual requires rescue and you are the Safety Attendant remember to stay calm and assess the situation. Check for any immediate dangers, such as falling debris or unstable structures, and any obstacles that could cause a problem during the rescue.
If the rescue is from water and possibly ICE. Do Not jump into the water and or venture out onto the ice. Use the lifeline to start placing tension on the line, reeling your co-worker into shore.
If the rescue is from a tank, tub, platform or other do not enter the area. Use the Lifeline to start placing tension on the line reeling your co-worker to the exit.
NOTE: If the co-worker is disabled, unconscious, you as the rescuer must wait for help. A planned rescue must be implemented.
It is important to remember that the primary role of the Safety Attendant is to monitor and ensure the safety of the worker. Depending on the immediate situation the Safety Attendant my be able to easily assist and or extract the co worker from the location.

The Safety Attendant as soon as they can, should always call for help.

COMMUNICATION: Immediately contact the designated rescue team or emergency services following emergency procedures. Be sure to provide them with as much information about the situation, including the nature of the emergency, the condition of the trapped person, and any specific hazards relating to the situation.

Maintain communication with the trapped individual if possible. Provide reassurance and guidance on actions they can take to protect themselves until the rescue team arrives.

ASCEND OR DESCEND: Depending on the situation, decide with the individual whether an extraction, ascent or descent is possible. Provide any information pertinent to the situation to emergency personnel and document it as well. Provide support to the individual in distress by staying in contact with them via agreed upon communication methods.
Do not enter any space or situation unless specifically trained to do so, as you yourself could become trapped.

RESCUE TECHNIQUES: The Safety Attendant's primary role is to monitor and ensure the safety of workers, not to perform rescues. The Safety Attendant must decide if they can assist the co-worker in their self rescue. If the trapped co-worker can not rescue them selves, one person trying a rescue of a co-worker will be almost impossible putting to risk the safety of the Safety Attendant.
Some of the things a Safety Attendant can do in a rescue situation that will make the process easier include initiating rescue protocol, assisting the rescue team if needed, and ensuring that any bystanders and non-essential personnel are kept away from the entry point to make sure there is a clear path for rescuers.

CALL FOR ASSISTANCE: Stay in contact with the designated rescue team or emergency services as per emergency procedures. Provide them with all necessary information about the situation including the condition of the individual in distress and any hazards present.

WAIT FOR ASSITANCE: While waiting for emergency assistance to arrive, remember to stay calm and alert. Secure the area to prevent further harm, if possible, without jeopardizing your own safety. This could include shutting down machinery, isolating energy sources, or cordoning off hazardous areas.
If you are trained in first aid and it safe to do so provide necessary initial care (control bleeding, treating for shock, CPR). Remember to communicate clearly and concisely with everyone (emergency personnel, trapped/injured persons, bystanders, etc.) and keep detailed records of everything that has occurred, and the actions taken. This information could be vital to emergency services once they arrive.

PLANNING & PRACTISING

Planning and practising a Rope Access procedure and a rescue is required. The simulation is to occur at the start of each new job and or shift change.

POST INCIDENT

STOP WORK, Report to your Supervisor, all incidents must be reported and no work can continue until FULL CLEARANCE and a RETURN TO WORK Instruction has been provided .

PPE	AB OHS Code Part 18 Section 228 (1) to 228 (3)
Jackets Flotation devises	AB OHS Code Part 18 Section 241 (1) to 241 (3)
Rope Access	AB OHS Code Part 41, Section 805 - 849
Work Alone	AB OHS Act, Section 2, 35, and 36
	AB OHS Code Alberta Regulation 191/2021 Work Alone Part 28
Right to Refuse	AB OHS Act Part 3 Section 35
Hazard Prevention	CA Part XIX Hazard Prevention Program 19.1

If you any questions STOP WORK and contact your supervisor.